

EECS 118 HW1

Devin Hua

1. Task 1 — Environment Setup

1.1 Introduction

- This section documents the setup of the Python development environment used for EECS 118 Homework 1. Python, Anaconda, Pandas, and Matplotlib were installed and tested successfully.

1.2 Verify Python Version and Running Python Interpreter

- `python --version`

```
C:\Users\huade\Desktop\EECS 118\hw1>python
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
Ctrl click to launch VS Code Native REPL (https://aka.ms/python-native-repl)
>>> print("hello world")
hello world
>>> 
```

Figure 1: Python version and “hello world” execution

1.3 Anaconda Installation

- Anaconda was installed following the official installation instructions.

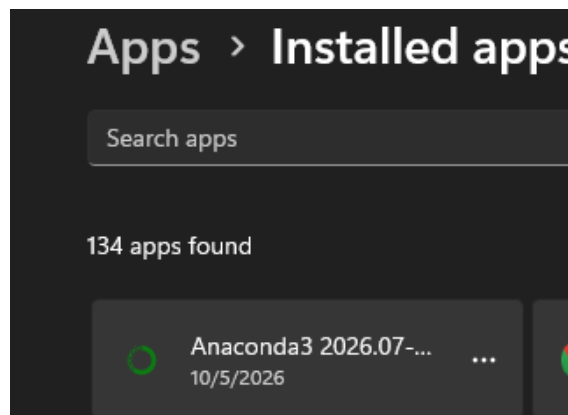


Figure 2: Anaconda installation

1.4 Creating the EECS 118 Environment

- Command: `conda create -n eeecs118 python=3.8`

```
(base) C:\Users\huade>conda create -n eeecs118 python=3.8
3 channel Terms of Service accepted
Retrieving notices: done
Channels:
- defaults
Platform: win-64
Collecting package metadata (repodata.json): done
Solving environment: done

==> WARNING: A newer version of conda exists. <==
current version: 26.5.3
latest version: 26.9.1

Please update conda by running

    $ conda self update

## Package Plan ##
```

Figure 3: Conda Create

1.5 Activating the Environment and Verify Python 3.8

- Command: `conda activate eeecs118`

```
(base) C:\Users\huade>conda activate eeecs118

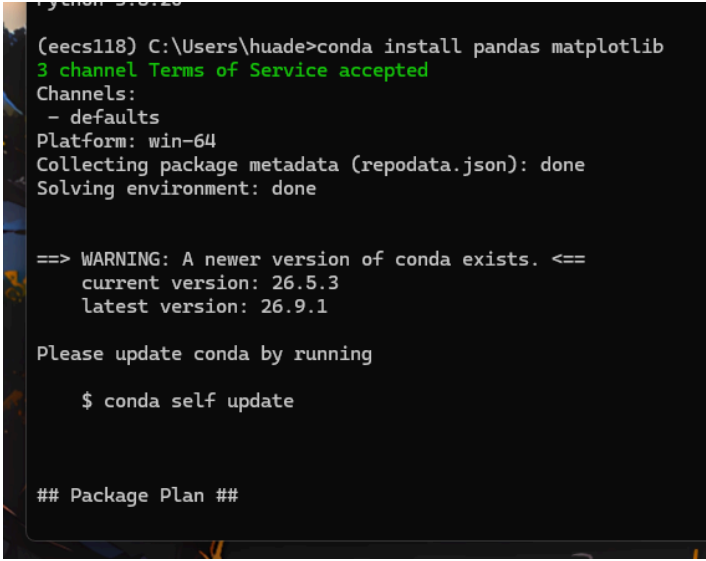
(eeecs118) C:\Users\huade>python --version
Python 3.8.20

(eeecs118) C:\Users\huade>
```

Figure 4: Activating Environment

1.6 Installing Required Packages

- Installation:
 - Pandas
 - Matplotlib

A terminal window with a dark background and light-colored text. The prompt is '(eecs118) C:\Users\huade>'. The command 'conda install pandas matplotlib' has been entered. The output shows '3 channel Terms of Service accepted', 'Channels: - defaults', 'Platform: win-64', 'Collecting package metadata (repodata.json): done', and 'Solving environment: done'. A warning message follows: '==> WARNING: A newer version of conda exists. <==', 'current version: 26.5.3', 'latest version: 26.9.1'. It then says 'Please update conda by running' followed by '\$ conda self update'. At the bottom, it says '## Package Plan ##'.

```
(eecs118) C:\Users\huade>conda install pandas matplotlib
3 channel Terms of Service accepted
Channels:
- defaults
Platform: win-64
Collecting package metadata (repodata.json): done
Solving environment: done

==> WARNING: A newer version of conda exists. <==
current version: 26.5.3
latest version: 26.9.1

Please update conda by running

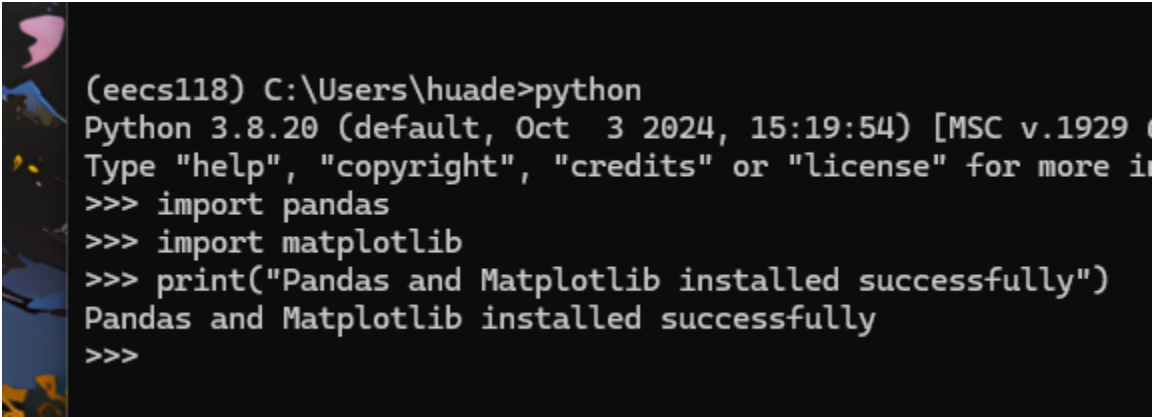
$ conda self update

## Package Plan ##
```

Figure 5: Package Installation

1.7 Testing Pandas and Matplotlib

- packages can be imported successfully:

A terminal window with a dark background and light-colored text. The prompt is '(eecs118) C:\Users\huade>'. The command 'python' has been entered. The output shows 'Python 3.8.20 (default, Oct 3 2024, 15:19:54) [MSC v.1929]', 'Type "help", "copyright", "credits" or "license" for more info', and the prompt '>>>'. The commands 'import pandas' and 'import matplotlib' have been entered. The command 'print("Pandas and Matplotlib installed successfully")' has been entered, and the output is 'Pandas and Matplotlib installed successfully'. The prompt '>>>' is shown again.

```
(eecs118) C:\Users\huade>python
Python 3.8.20 (default, Oct 3 2024, 15:19:54) [MSC v.1929]
Type "help", "copyright", "credits" or "license" for more info
>>> import pandas
>>> import matplotlib
>>> print("Pandas and Matplotlib installed successfully")
Pandas and Matplotlib installed successfully
>>>
```

Figure 6: Testing Pandas and Matplotlib

1.8 Environment Setup Summary

The Python environment for EECS 118 Homework 1 was successfully set up. Python was first installed and verified by checking the Python version and running a simple “Hello World” program. An Anaconda environment named `eeecs118` was then created using Python 3.8.20. The required Pandas and Matplotlib packages were installed and successfully imported within the environment. The environment is now ready for performing the dataset inspection and data visualization tasks in Homework 1.

Section 2 — Task 2.1: Dataset Inspection

2.1 Score Dataset (`score.csv`)

2.1.1 Number of Data Instances

- How many data instances (students) exist in this dataset?

```
Q1: Data instances (students) in dataset: 25
```

- The score dataset contains **25 data instances**, with each instance representing one student. This was determined by finding the number of rows in the Pandas DataFrame. Therefore, the dataset contains study-hour and score information for 25 students.

2.1.2 Number of Columns and Data Types

- How many columns are there in this dataset? What data type is each of them?

```
Q2: Columns in dataset: 2
Data types of each column: Hours    float64
Scores    int64
dtype: object
```

- The dataset contains **2 columns: Hours and Scores**. Both columns contain numerical data. The `Hours` column represents the number of hours a student studied, while the `Scores` column represents the student's score.
- The data types were obtained using the Pandas `dtypes` property. Since both columns contain numerical values, they can be used directly for mathematical calculations and data visualization.

2.1.3 Average Study Hours

- What is the average number of study hours in this dataset?

```
Q3: Average Study Hours: 5.012
```

- The average number of study hours is approximately **5.01 hours**. This value was calculated by adding all values in the Hours column and dividing by the number of students.
- This result indicates that the students in the dataset studied for approximately five hours on average.

2.1.4 Average Score

- What is the average number of scores in this dataset?

```
Q4: Average Scores: 51.48
```

- The average score is approximately **51.48**. This was calculated by finding the mean of the Scores column.
- The average provides an overall measure of student performance in the dataset. However, it does not show how the scores are distributed among individual students, which is why the visualization in Task 2.2 is useful.

2.1.5 Observation of the Score Dataset

- This dataset can be used to analyze the relationship between study hours and scores. It can help identify if there is a correlation between the amount of time spent studying and the scores achieved by students.

2.2 Cereal Dataset (cereal.csv)

2.2.1 Number of Cereal Products

- How many data instances (cereal products) exist in this dataset?

```
Q1: Data instances (cereals) in dataset: 77
```

- The cereal dataset contains **[number] cereal products**. Each row represents one cereal product and contains information about characteristics such as its manufacturer, calories, sugars, fiber, and rating.

- This allows the dataset to be used to compare the nutritional characteristics and ratings of different cereal products.

2.2.2 Data Types of Each Column

- What are the types of each dataset column?

```
Q2: Data types of each column: name      object
mfr      object
type      object
calories  int64
protein   int64
fat        int64
sodium    int64
fiber      float64
carbo      float64
sugars     int64
potass     int64
vitamins   int64
shelf      int64
weight     float64
cups       float64
rating     float64
dtype: object
```

- The dataset contains both categorical and numerical columns. The mfr column is categorical because it contains manufacturer abbreviations, while columns such as calories, sugars, fiber, and rating contain numerical values.
- The data types were determined using Pandas' dtypes property. Knowing the data type of each column is important because different types of analysis can be performed on categorical and numerical data.

2.2.3 Number of Cereal Manufacturers

- How many cereal manufacturers (mfr) are there in the dataset?

```
Q3: Unique manufacturers in dataset: 7
```

- There are **7 unique manufacturers** represented in the dataset. This was determined using the unique() function on the mfr column and counting the resulting values.
- This shows that the dataset contains cereal products from multiple manufacturers, allowing their products to be compared.

2.2.4 Maximum, Minimum, and Average Values

- What is the max/min/average value for each column in the dataset?

Q4: Max/Min/Average for each column:

	calories	protein	fat	sodium	fiber	carbo	sugars	potass	vitamins	shelf	weight	cups	rating
count	77.000000	77.000000	77.000000	77.000000	77.000000	77.000000	77.000000	77.000000	77.000000	77.000000	77.000000	77.000000	77.000000
mean	106.883117	2.545455	1.012987	159.675325	2.151948	14.597403	6.922078	96.077922	28.246753	2.207792	1.029610	0.821039	42.665705
std	19.484119	1.094790	1.006473	83.832295	2.383364	4.278956	4.444885	71.286813	22.342523	0.832524	0.150477	0.232716	14.047289
min	50.000000	1.000000	0.000000	0.000000	0.000000	-1.000000	-1.000000	-1.000000	0.000000	1.000000	0.500000	0.250000	18.042851
25%	100.000000	2.000000	0.000000	130.000000	1.000000	12.000000	3.000000	40.000000	25.000000	1.000000	1.000000	0.670000	33.174094
50%	110.000000	3.000000	1.000000	180.000000	2.000000	14.000000	7.000000	90.000000	25.000000	2.000000	1.000000	0.750000	40.400208
75%	110.000000	3.000000	2.000000	210.000000	3.000000	17.000000	11.000000	120.000000	25.000000	3.000000	1.000000	1.000000	50.828392
max	160.000000	6.000000	5.000000	320.000000	14.000000	23.000000	15.000000	330.000000	100.000000	3.000000	1.500000	1.500000	93.704912

- The maximum, minimum, and average values were calculated using Pandas' `describe()` function. These statistics summarize the numerical columns in the dataset.
- The results show the range of values for nutritional characteristics and ratings. For example, the minimum and maximum values show how much a nutritional characteristic varies between cereal products, while the mean represents its average value across the dataset.
- These statistics provide a general overview of the nutritional content and ratings of the cereals before examining the relationships between individual variables in Task 2.2.

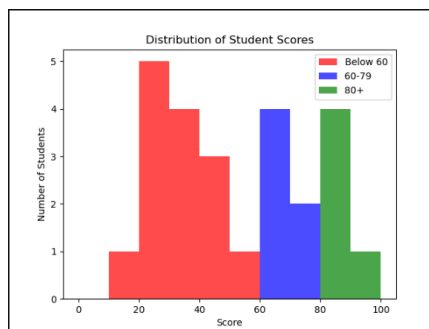
2.2.5 Observation of the Cereal Dataset

- This dataset can be used to analyze the nutritional content of different cereals and their manufacturers. It can help identify trends in cereal production and consumer preferences based on nutritional values.

Section 3 — Task 2.2: Data Visualization

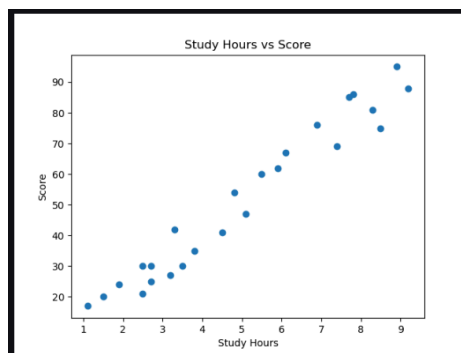
3.1 Score Dataset (score.csv)

3.1.1 Score Distribution Histogram



- The histogram shows the distribution of student scores across three score ranges: scores below 60 are shown in red, scores from 60 to below 80 are shown in blue, and scores of 80 or higher are shown in green.
- The results show that most students received scores below 60, while fewer students scored between 60 and 80 or above 80. There are only a small number of students with scores of 80 or higher.
- Overall, the histogram indicates that the scores are concentrated more heavily in the lower score range. This provides a clearer view of the score distribution than the average score alone because it shows how the individual scores are spread across different ranges.

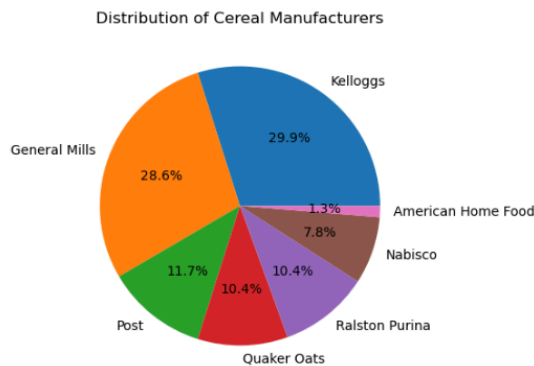
3.1.2 Study Hours vs. Score Scatterplot



- The scatterplot shows the relationship between the number of hours a student studied and their score. Study hours are plotted on the x-axis, while scores are plotted on the y-axis.
- The results show a general positive relationship between study hours and scores. In other words, students who studied for more hours generally achieved higher scores. Although the points do not form a perfectly straight line, the overall trend indicates that increased study time is associated with better scores.
- This suggests that study time may be an important factor related to student performance in this dataset.

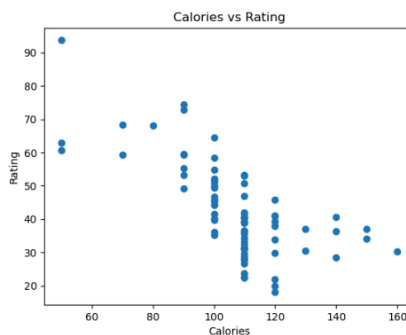
3.2 Cereal Dataset (cereal.csv)

3.2.1 Cereal Manufacturer Distribution



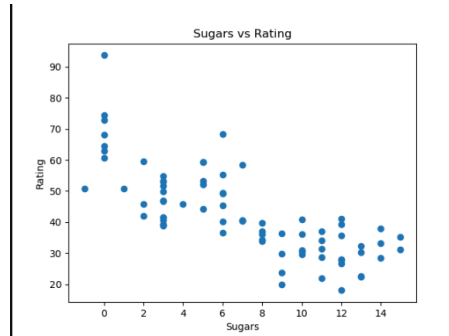
- The pie chart shows the distribution of cereal products among the different manufacturers in the dataset. Each section represents a manufacturer, and the size of each section corresponds to the number of cereal products produced by that manufacturer in the dataset.
- The results show that some manufacturers have a larger representation than others. The manufacturer with the largest section has the most cereal products in the dataset, while manufacturers with smaller sections have fewer products.
- This chart provides a quick way to compare the representation of different manufacturers in the dataset.

3.2.2 Calories vs. Rating



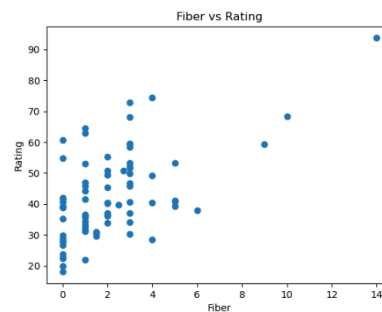
- The scatterplot shows the relationship between the number of calories in a cereal and its rating. Calories are plotted on the x-axis and rating is plotted on the y-axis.
- The points do not appear to form a strong upward or downward pattern. This suggests that the number of calories alone does not have a clear relationship with cereal rating in this dataset.
- Therefore, calories may not be a strong predictor of how highly a cereal is rated.

3.2.3 Sugars vs. Rating



- The scatterplot shows the relationship between the amount of sugar in a cereal and its rating. Sugar content is plotted on the x-axis and rating is plotted on the y-axis.
- The results show a general downward trend, with cereals containing higher amounts of sugar tending to have lower ratings. However, the points are not perfectly aligned, meaning that sugar content is not the only factor affecting cereal ratings.
- Overall, the visualization suggests that cereals with lower sugar content tend to receive higher ratings in this dataset.

3.2.4 Fiber vs. Rating



- The scatterplot shows the relationship between the amount of fiber in a cereal and its rating. Fiber content is plotted on the x-axis and rating is plotted on the y-axis.
- The results show a general positive relationship between fiber and rating. Cereals with higher fiber content tend to have higher ratings, although there are some exceptions.
- This suggests that fiber content may be positively associated with cereal ratings in this dataset.